

Date: Wednesday, July 2, 2025 5:34:34PM

PQ Controls INC.

Page 1

Indented Bill of Material NS

From Item No. [312S105-006] To [312S105-006]

Assembly: 312S105-006 Rev: 0

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
1	B-15140	HOUSING ASSY 312, 2 AXIS NO ON-AXIS DETENT	1.0000	EA
2	C-14943	312, HOUSING, MERCURY MARINE V2	1.0000	EA
2	B-15194	BRASS INSERT, 112/212 HOUSING	1.0000	EA
2	A-09354	GREASE DOW CORNING 33	0.0000	A/R
2	B-10252	STAMPING, RETAINER, BALL M212	1.0000	EA
2	A-06936-0.50S	SCREW #6 X 1/2 PLASTITE 48-2 PAN HD SS TAP	4.0000	EA
2	B-14958	312, SHAFT WIRE STRAIN RELIEF	1.0000	EA
2	A-00600-04	WIRE TIE 4" BLACK 18 LBS PLT1M-M0	2.0000	EA
2	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	B-15145-R	LED RING ASSY. 312, REMOTE / JPI / JPID	1.0000	EA
2	A-14901-01	ELEC ASSY., LED RING MERCURY MARINE	1.0000	EA
3	A-14901	SMT ASSY, LED BOARD (312 JOYSTICK)	1.0000	EA
4	B-14900	PCB LED Ring for Mercury Starfish Joystick	1.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	2.0000	EA
4	SZ-15157	C1,2 FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	7.0000	EA
4	SR-11617-0000	F1,3,4,5,6,7,8 1/10W 0 OHM SURFACE (0603)	1.0000	EA
4	SD-15069	F2 SMD LED L955H-LOLPGQBC TRI-COLOR	32.0000	EA
4	SR-12545-4990	LED1-16, 21-36 1/10W 1% 499 OHM THICK FILM (0805)	2.0000	EA
4	SR-12545-1501	R1,2 1.5K OHM 1/10W 1% THICK FILM	3.0000	EA
4	SU-15162	R3,4,5 TLC59116IPWR (TI) LED Lighting Driver	2.0000	EA
3	A-15059-08	U1,2 ACC-437-ND 8 conductor flex cable	1.0000	EA
2	C-14949-01	312, CHROME SURROUND (REMOTE), MERCURY MARINE V2 *	1.0000	EA
2	B-14950	LIGHT PIPE, LED RING MERCURY V2, HARDCOATED *	1.0000	EA
2	A-15252	PRIMER, DOW CORNING PR-1205 *	0.0000	A/R

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 2

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
2	A-15251	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER KIT *	0.0000	A/R
3	A-15251-A	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER BASE	0.0000	A/R
3	A-15251-B	RTV SILICONE, DOW CORNING SYLGARD 184 ELASTOMER CURING AGENT	0.0000	A/R
2	A-11285-0.375	SCREW #2 X 3/8" PLASTITE 48-2 SELF TAPPING	3.0000	EA
2	A-15060-08	487545-5 8 pos Connector plug	1.0000	EA
2	A-15282	ASSEMBLY PROCEDURE, 312 POTTING, MERCURY MARINE	0.0000	
1	B-15054-104	SHAFT ASSY 312 FOR TKG2	1.0000	EA
2	B-14957	SHAFT, 312 TKG2	1.0000	
2	B-10088	BALL, M212	1.0000	EA
2	A-10498-03	PIN GROOVED 5/32 X 1/2 TYPE E LOW CARBON	1.0000	EA
2	A-08113-609	LOCTITE 609 THREADLOCKER GREEN	0.0000	A/R
2	B-14945	312, X/Y AXIS ENCAPSULATED MAGNET HOLDER *	1.0000	EA
3	B-14830	MAGNET, MLX90333 SMT HALL EFFECT	1.0000	EA
2	A-00780-08SS	ROLL PIN, 1/16 X 1/2 STAINLESS STEEL	1.0000	EA
2	A-11050	ASSEMBLY PROCEDURE M212 SHAFT ASSY	1.0000	
1	B-15141	CENTERING ASSY, 312	1.0000	EA
2	C-10084	M212 CENTERING FOOT JLG#: 7020555	1.0000	EA
2	B-13471	SPRING, CENTERING LIGHT 112/212	1.0000	EA
2	B-14953	SLEEVE, 312	1.0000	EA
2	A-10500-37SS	RETAINING RING, STAINLESS STEEL	1.0000	EA
2	B-14952	BOOT MOLDING 312	1.0000	EA
2	A-00451-09	O-RING 3/8" NITRILE DURO 2-012	1.0000	EA
2	A-05447-S-316	WASHER FLAT 5/8" X 3/8", ***316 STAINLESS STEEL	1.0000	EA
2	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	C-15142-R	GRIP ASSEMBLY TKG2 TWIST, REMOTE / JPI / JPID / SNOOK	1.0000	EA
2	B-14978	TOP GRIP HALF V2, CHROME PLATED	1.0000	EA
3	D-14939	TOP HALF, TWIST KNOB GRIP V2, CASTING	0.0000	EA
2	B-14955	312, PIVOT BLOCK W/ENCAPSULATED MAGNET *	1.0000	EA
3	B-14830	MAGNET, MLX90333 SMT HALL EFFECT	1.0000	EA
2	B-13472	TWIST ARMS, TWIST KNOB GRIP	2.0000	EA

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 3

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
2	A-13620-01	BEARING, OIL FILLED .378 ID X .469 OD X .304 H *	1.0000	EA
2	B-13491-SS	SPRING, EXTENSION, TWIST KNOB STAINLESS STEEL	1.0000	EA
2	B-14979	BOTTOM GRIP HALF V2, CHROME PLATED	1.0000	EA
3	C-14940	BOTTOM HALF, TWIST KNOB GRIP V2, CASTING	0.0000	EA
2	B-14951	GRIP SLEEVE TWIST KNOB V2	1.0000	
2	A-00703-03	SCREW 2-56 X 3/16 PHIL PAN MS STAINLESS STEEL	3.0000	EA
2	A-13890	ADHESIVE, DOUBLE/BUBBLE URETHANE	0.0000	A/R
2	A-00161-02S	LOCK WASHER #2 INTERNAL TOOTH STAINLESS STEEL	3.0000	EA
2	A-10498-04	PIN GROOVED 5/32 X 5/8 TYPE E LOW CARBON	1.0000	EA
2	A-15002-06S	4-40 X 3/4" BUTTON HEAD SOCKET CAP SS	2.0000	EA
2	A-00451-02	O-RING, TWIST KNOB GRIP V2, ID = 1.669	1.0000	EA
2	A-01349-13	ROLL PIN 1/8 X 1" ZINC YELLOW	1.0000	EA
2	A-02686-03S	RING, RETAINING 3/8" SS	1.0000	EA
2	B-15082-01	MERCURY MARINE JOYSTICK V2, KNOB DISPLAY ASSY., REMOTE/JPI *	1.0000	EA
3	A-15783	KNOB DISPLAY APPEARANCE GUIDE	0.0000	
2	A-15080-02	ELEC ASSY, MERCURY V2 KNOB, REMOTE VERSION	1.0000	EA
3	A-14851-REM	SMT ASSY, KNOB BOARD (312 JOYSTICK, REMOTE VERSION)	1.0000	EA
4	B-14850	PCB Knob PCB for Mercury Starfish Joystick	1.0000	EA
4	SC-10228-01	TAJA106K010RNJ CAPACITOR TANT 10UF 10V SURFACE MOUNT	2.0000	EA
4	SC-14635-02	C2,3 0.01uf 50V 0603	5.0000	EA
4	SC-14635-03	C4,8,10,12,17 C0603C104K4RACTU 0.1uf 16V 0603 Bond	7.0000	EA
4	SC-12564-05	C5,7,9,11,13,16,18 4.7UF, 16V CAP CERAMIC (0805)	1.0000	EA
4	SC-14907-01	C6 18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond	2.0000	EA
4	SZ-15157	C14,15 FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond	1.0000	EA
4	SR-11617-0000	FER1 1/10W 0 OHM SURFACE (0603)	3.0000	EA
4	SQ-14908	R23,24,25 MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.)	1.0000	EA

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 4

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
		Q1		
4	SR-12545-1002	NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM (0805)	3.0000	EA
		R1,2,7		
4	SR-12552-1200	1/10W 5% 120 OHM THICK FILM 0805	1.0000	EA
		R3		
4	SR-11617-1001	1/10W 1% 1K OHM SURFACE	1.0000	EA
		R4		
4	SR-12545-1501	1.5K OHM 1/10W 1% THICK FILM	1.0000	EA
		R6		
4	SU-15156	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-48) STM32F103C8T6	1.0000	EA
		U1		
4	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG	1.0000	EA
		U3		
4	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC	1.0000	EA
		U4		
4	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal	1.0000	EA
		Y1		
3	A-14857	SMT ASSY, KNOB SENSOR BOARD (312 JOYSTICK)	1.0000	EA
4	B-14856	PCB Knob Sensor PCB for Mercury Startfish Joystick	1.0000	EA
4	SU-15161	MLX90333LGO-BCH-100, 3D POSITION HALL SENSOR, DUAL DIE, SPI	1.0000	EA
4	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond	4.0000	EA
3	A-15062	'850-10-007-20-001101 PRECI-DIP 7 pos interconnect pins	1.0000	EA
3	A-12827-01-RD	RED WIRE, 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-12827-01-BK	BLACK WIRE 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-12827-01-BR	BROWN WIRE 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-12827-01-GY	GRAY WIRE, 26GA TEFLON TYPE E 19/38	0.5833	FT
3	A-15578	ADHESIVE, HOT MELT 3779	0.0000	A/R
2	A-15193	GREASE, CMD EXTREME PRESSURE LUBE #3	0.0000	A/R
1	A-14881-01	ELEC ASSY, MERCURY V2 MOTHERBOARD	1.0000	EA
2	A-14881	SMT ASSY, MOTHERBOARD BOARD (312 JOYSTICK)	1.0000	EA
3	B-14880	X/Y PCB for Mercury Starfish Joystick	1.0000	EA
3	SC-09917-01	CAPACITOR .01UF 200V CERAMIC Bond (1210)	2.0000	EA
		C1,59		

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 5

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
3	SC-09917-05	CERAMIC CAPACITOR 1uF 25V 10% X7R (1210) C5	1.0000	EA
3	SC-10228-01	TAJA106K010RNJ CAPACITOR TANT 10UF 10V SURFACE MOUNT C10	1.0000	EA
3	SC-14635-01	GRM1885C1H111JA01D 110pF 16V 5% CERAMIC CAP 0603 C11,12,13,14,33,34,35,36	8.0000	EA
3	SC-14635-02	0.01uf 50V 0603 C15,18,19,20,21,22,37	7.0000	EA
3	SC-14635-03	C0603C104K4RACTU 0.1uf 16V 0603 Bond C16,17,24,25,26,27,28,29,30,31,32,39,40,45,48	15.0000	EA
3	SC-12564-05	4.7UF, 16V CAP CERAMIC (0805) C23	1.0000	EA
3	SC-14907-01	18pF 16V 5% CERAMIC CAPACITOR, 0603 Bond C41,42	2.0000	EA
3	SC-12564-10	22uF, 20% 35V CAP CERAMIC (0805) C46	1.0000	EA
3	SC-12564-11	C2012X5R1V156M125AC 15uF, 20% 35V CAP C47,49,50,54,52,53	6.0000	EA
3	SC-12564-07	0.1UF, 50V, CAP CERAMIC (0805) C54	1.0000	EA
3	SC-14635-06	.0082uF 16V 10% CERAMIC CAP 0603 Bond C55	1.0000	EA
3	SC-09160-02	CL31A106KOHNNNE 10UF, 16V CAPACITOR, CERAMIC 10% (1206) C56,57	2.0000	EA
3	SC-09917-02	CAPACITOR 0.1UF 200V CERAMIC SMT (1210) C58,60,61	3.0000	EA
3	SD-15271	20V TVS DIODE BI-DIR. SMCJ20CA D1,3	2.0000	EA
3	SD-15272	B560C-13-F SCHOTTKY DIODE 60V 5A Bond D2	1.0000	EA
3	SD-12845	B250-13-F 2A SCHOTTKY DIODE D4	1.0000	EA
3	SD-15273	NUP2105LT1G TVS DIODE ARRAY 27V CANBUS PROTECTION ## D5,6	2.0000	EA
3	SD-15274	TVS DIODE 5Vr 400W UNI-DIR. (LITTELFUSE #SMAJ5.0A) ## D11,12,13	3.0000	EA

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 6

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

Component Item Number or Subassembly		Description / Comment	Qty Needed	Units
3	SD-15275	TVS DIODE ESD PROTECTION 5V SOD523 Bond ## D7,8,9,10,14,15,16,17,18	9.0000	EA
3	SZ-15157	FERRITE CHIP, 1K OHM 0.8A (0603 PACKAGE) MPZ1608S102A Bond FER1,2,3,4,5,6,7,8,9,11,12,13	12.0000	EA
3	SZ-15276	FERRITE BEAD 600 OHM 2A (0805) TDK FER14,15	2.0000	EA
3	SL-15277	CHOKE COM MODE 1mH 0.8A Bond L1	1.0000	EA
3	SL-14637	CHOKE COM MODE 2x51uH 0.3A 744212510 L2,3	2.0000	EA
3	SL-15278-01	INDUCTOR 27uH 30% SMD 6038 Bond L4	1.0000	EA
3	SL-15278-02	CDRH6D38NP-7R4NC INDUCTOR 7.4uH 30% SMD 6038 L5	1.0000	EA
3	SQ-14908	MOSFET N-CHANNEL 60V, BS170FTA (Diodes Inc.) Q1,2	2.0000	EA
3	SQ-15155	TRANSISTOR, NPN 100mA 50V MUN2211T1G Q4,5,6	3.0000	EA
3	SQ-15166	TRANSISTOR, PNP 50V DPLS350E-13 Q7,8,9	3.0000	EA
3	SR-12545-1200	120 OHM 1/10W 1% THICK FILM 0805 R1	1.0000	EA
3	SR-12545-1001	1K OHM 1/10W 1% THICK FILM R12,25,37,39,41	5.0000	EA
3	SR-12545-1002	NRC10F1002TRF 10K OHM 1/10W 1% THICK FILM (0805) R5,27,28	3.0000	EA
3	SR-12545-1501	1.5K OHM 1/10W 1% THICK FILM R6,7,8,9,10,11,21,26,29	9.0000	EA
3	SR-09152-3012	1/4W 1% 30.1K OHM SURFACE MOUN R31	1.0000	EA
3	SR-12545-3320	Resistor 332 ohm 1% 1/4W 0805 R32,33,38,40,42	5.0000	EA
3	SR-09152-2003	1/4W 1% 200K OHM SURFACE MT R36	1.0000	EA
3	SR-12545-0000	0 OHM 1/10W 1% THICK FILM (0805) R46,47	2.0000	EA

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 7

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
3	SR-09152-01.10	1/4W 1% 1.1 OHM SMT R48	1.0000	EA
3	SR-15279-2002	1/4W 1% 20K OHM SMT 0805 PACKAGE R49	1.0000	EA
3	SR-15281-6802	1/2W 1% 68K OHM ANTI SURGE SMT 0805 PACKAGE R50	1.0000	EA
3	SR-15281-1202	1/2W 1% 12K OHM ANTI SURGE SMT 0805 PACKAGE R51	1.0000	EA
3	SU-15169	32 BIT ARM MICROCONTROLLER, 72MHz (LQFP-64) STM32F105R8T6 U1	1.0000	EA
3	SU-15280	REGULATOR, SWITCHING 40V 2A U2	1.0000	EA
3	SU-12791-3.3BS	MIC5209-3.3YSTR 500MA LO-NOISE LDO REG U4	1.0000	EA
3	SU-14405	SN65HVD1050AQDRQ1 CAN TRANSCEIVER W/HIGH SPEED & EMC U5,8	2.0000	EA
3	SU-15161	MLX90333LGO-BCH-100, 3D POSITION HALL SENSOR, DUAL DIE, SPI U7	1.0000	EA
3	SY-14550-08	ABM3-8.000MHZ-D2Y-T 8MHz 18pF miniature SMT Crystal Y1	1.0000	EA
2	A-15055-04	5-103669-3 4 pos Connector Header	1.0000	EA
2	A-15058-08	1-104071-0 8 pos Connector Header for Mercury x/y board	1.0000	EA
2	A-15058-05	5-104071-2 5 pos connector header	1.0000	EA
2	A-06977-36	HEADER BERG 36 POS 68000-436 STRAIGHT	0.0000	EA
2	A-07794-01	CONFORMAL COATING #MS-470C SPRAY	0.0000	A/R
2	A-15083	PRETEST PROCEDURE, 312 MERCURY MARINE	0.0000	
2	A-15185	SOFTWARE FLASHING PROCEDURE, 312 MERCURY MARINE	0.0000	
1	B-15144	ENCLOSURE ASSY. 312	1.0000	EA
2	C-14944	312, ENCLOSURE	1.0000	EA
2	A-00451-13	O-RING, 312 JOYSTICK	1.0000	EA
2	A-06936-0.87S	SCREW #6 X 7/8 PLASTITE 48-2 PAN HD SS	4.0000	EA
2	A-13718-01	FLAT RUBBER WASHER .140 X .312 X 1/32	4.0000	EA
2	A-03485-01	LIQUID TIGHT STRAIN RELIEF HEYCO 3207 SM	1.0000	EA

Date: Wednesday, July 2, 2025 5:34:34PM

Indented Bill of Material NS

Assembly: 312S105-006 Rev: 0

PQ Controls INC.

Page 8

Description: 3 AXIS TKG2 CANBUS B-15299* JPI DUAL VERSION

<u>Component Item Number or Subassembly</u>		<u>Description / Comment</u>	<u>Qty Needed</u>	<u>Units</u>
2	B-14956	312 MOUNTING BRACKET W/STUDS ASSY, MERCURY MARINE	1.0000	EA
2	A-01261-04S	LOCK NUT NYLON 4-40 STAINLESS STEEL	3.0000	EA
2	A-15306	HYDROPHOBIC VENT PATCH, LTI ATLANTA #074-1000-000	1.0000	EA
1	B-15081-JPID	KEYPAD ASSY., MERCURY MARINE STARFISH JOYSTICK *	1.0000	EA
2	A-16228	KEYPAD PRETEST PROCEDURE, 312 MERCURY MARINE	0.0000	
1	A-11285-0.25S	#2 X 1/4" PLASTITE SELF TAPPING 48-2 SS	2.0000	EA
1	A-11285-0.375S	SCREW #2 X 3/8" PLASTITE 48-2 SS	2.0000	EA
1	A-14954	8M0111425 NEMA cable with orange heat shrink	1.0000	EA
1	A-15056-04	104257-3 4 pos Connector plug	1.0000	EA
1	A-15057	SOCKET, REEL MFG # 1-104480-2, Sockets for 104257-3 p	4.0000	EA
1	B-15079-02	312 BEZEL ASSY., REMOTE/JPID/SNOOK, MERCURY MARINE V2	1.0000	EA
2	C-14946-01	312 BEZEL, UPPER (REMOTE), MERCURY MARINE V2	1.0000	EA
2	B-14948	312, CHROME BEZEL TRIM, MERCURY MARINE V2 *	1.0000	EA
2	C-14947	312 BEZEL, LOWER, MERCURY MARINE V2	1.0000	EA
2	A-15004	DECAL, MERCURY JEWEL P/N 37-8M6001063	1.0000	EA
1	B-14941	GASKET 312, EPDM SPONGE	1.0000	EA
1	A-15230	# 10-6957 SCREW #10-16 X 3/4 HI-LO RECESSED PAN HD	4.0000	EA
1	A-15034	ASSEMBLY PROCEDURE 312 MERCURY MARINE V2	0.0000	
1	S1145	SOFTWARE, MERCURY MARINE 312S105-006 JPI DUAL	0.0000	

1 bill of material printed